

Macroeconomic Theory and Policy: Lecture W1 (Ch.7 and Ch.8)

What are You Going to Learn (Ch. 7)

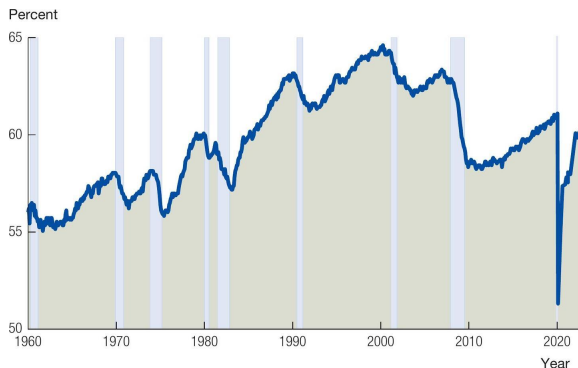
- Some fact about labour data
- How labor market distortions like taxes and firing costs affect employment in the long run

The U.S. Labor Market

- In the U.S. labor market (similar for Canada),
 - ▶ Wages account for two-thirds of per capita GDP
 - ▶ Average wages have grown 2 percent per year for the last century
 - ★ Roughly the same as the GDP growth
 - ★ But, not all are growing at the same rate
- The employment-population ratio:
 - ▶ Fraction of the civilian population over the age of 16 that is working
 - ▶ Has been increasing over time
 - ▶ Decreases during recessions

Ratio of Employment to Population in the United States, 1960–2016

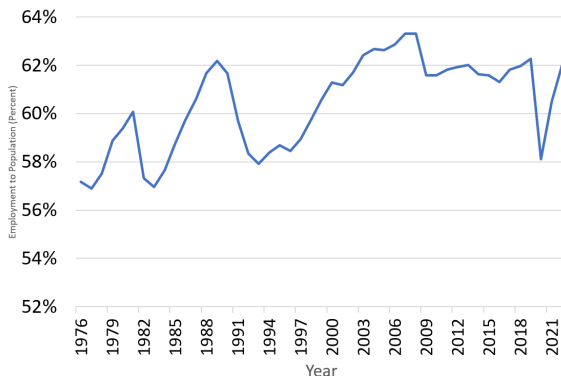
- Over the last 50 years, the employment-population ratio has been increasing (peak around 65%)
 - ▶ Mostly due to the increase in female workers participation
- Decline during recession in particular during 2007-2009 and other recession



Source: U.S. Bureau of Labor Statistics, Civilian Employment-Population Ratio [EMRATIO], retrieved from FRED, Federal Reserve Bank of St. Louis; fred.stlouisfed.org/series/EMRATIO. For this statistic, the population is the civilian, noninstitutionalized population at least 16 years of age.

Ratio of Employment to Population in the Canada, 1976–2021

- Canada's number is similar to that of the US
- The notable difference is that the ratio was still rising during the US recession in year 2000



⁰Source: Organization for Economic Co-operation and Development, Employed Population: Aged 15 and over: All Persons for Canada [LFEMTTTTTCAM647S], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/LFEMTTTTTCAM647S>.

The Composition of the U.S. Labor Force, February 2016

- A person is unemployed if (i) does not have a job, (ii) has actively searched for a job in the last 4 weeks, and (iii) Is available for work

TABLE 7.1

The Composition of the U.S. Labor Force, January 2019

Civilian population, aged 16 and over	258 million
Labor force	163 million
Employed	157 million
Unemployed	6.5 million
Not in the labor force	95 million

Source: Bureau of Labor Statistics, Economic News Release, February 1, 2019.

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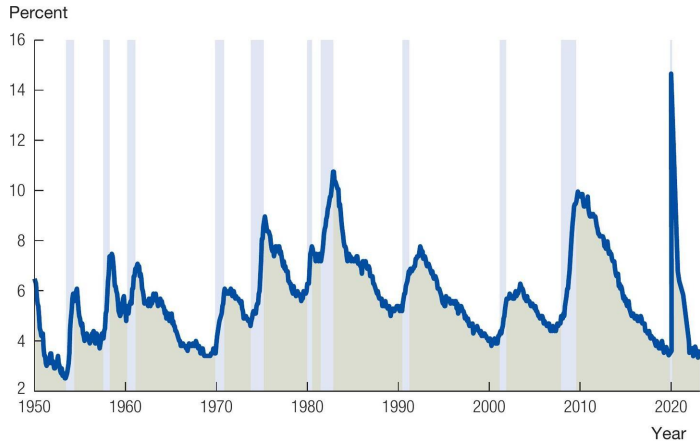
The U.S. Labour Market

- The unemployment rate: the fraction of the labor force that is unemployed

$$\text{Unemployment Rate} = \frac{\text{Unemployed}}{\text{Labour Force}} \times 100$$

- The unemployment rate is another common measure for analyzing the state of the labor market
- There are different types of unemployment: Frictional, Structural and Cyclical
 - ▶ We will cover them later
- There are a few alternative measure of unemployment, but most of them are highly correlated

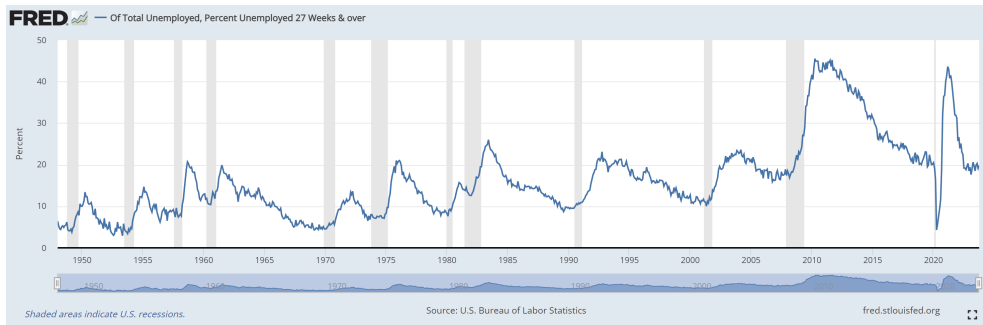
The U.S. Unemployment Rate



Source: U.S. Bureau of Labor Statistics, Civilian Unemployment Rate [UNRATE],
retrieved from FRED, Federal Reserve Bank of St. Louis;
fred.stlouisfed.org/series/UNRATE.

The Dynamics of the Labor Market

- For most people, periods of unemployment are relatively short
 - ▶ Each month, about 25 percent of people who are unemployed will find a job
- However, a significant fraction remain unemployed for long periods of time



- Job creation and job destruction occur each month
 - ▶ Some workers would be hired, while some quit their job or got fired
 - ▶ Happen during both expansionary period or recession

Different Kinds of Unemployment

- Cyclical unemployment: Associated with short-run fluctuations in output
 - ▶ Due to business cycle
 - ▶ Measure as the difference between actual rate and natural rate
- The natural rate of unemployment: Rate that would prevail with no cyclical unemployment
 - ▶ Frictional unemployment: Workers between jobs in the dynamic economy
 - ★ e.g., Instead of accepting the first available job, workers take time to match their skills with demands of firms, etc
 - ▶ Structural unemployment: labour market failure to match workers and firms
 - ★ e.g., Hiring and firing costs, the level of unemployment benefits, the level of the minimum wage, etc

Bathtub Model of Unemployment

- Recall that job separation and matching happen at the same time
- Bathtub model: States how employment and unemployment evolve over time

$$E_t + U_t = \bar{L}$$

where

E_t : Employment

U_t : Unemployment

$\bar{L}$: Size of Labour Force

- E_t and U_t are endogenous variables while $\bar{L}$ is a constant

Bathtub Model of Unemployment

- Unemployment changes over time by following:

$$\Delta U_{t+1} = \bar{s}E_t - \bar{f}U_t$$

where

ΔU_{t+1} is the change in employment

$\bar{s}$ is the job separation rate

$\bar{f}$ is the job-finding rate

- The term $\bar{s}E_t$ is the number of employed people who lose their jobs
- The term $\bar{f}U_t$ is the number of unemployed people who find new jobs
- The value for $\bar{s}$ is usually around value 1.4% per month

Bathtub Model of Unemployment

- The bathtub model has a steady state – the number of people employed and unemployed is constant
 - ▶ Occurs when the change in unemployment is zero
 - ▶ The number of people losing jobs exactly equals the number of people finding new jobs
- Set the change in unemployment (ΔU_{t+1}) to zero and sub in the unemployment def

$$0 = \bar{s}E_t - \bar{f}U_t = \bar{s}(\bar{L} - U_t) - \bar{f}U_t = \bar{s}\bar{L} - (\bar{s} + \bar{f})U_t$$

- Solving for U_t

$$U_t = \frac{\bar{s}\bar{L}}{\bar{s} + \bar{f}}$$

Bathtub Model of Unemployment

- The unemployment rate is then

$$u^* = \frac{U^*}{\bar{L}} = \frac{\bar{s}}{\bar{s} + \bar{f}}$$

- This unemployment rate holds in the steady state (i.e., long run), this would be the natural rate of unemployment in the Bathtub Model
- To alter the natural rate of unemployment:
 - ▶ Change the job-finding rate, $\bar{f}$
 - ▶ Change the job separation rate, $\bar{s}$

Bathtub Model of Unemployment

- Some policies may have alter the unemployment rate. For example, one may try to reduce the job separation rate by imposing firing costs on firms:
 - ▶ Any firm that fires a worker must pay one year's salary as a severance payment.
 - ▶ Firms may then become reluctant to hire new workers, reducing the job-finding rate
 - ▶ The net of these effects could be to increase the natural rate of unemployment rather than to decrease it

Bathtub Model of Unemployment

- What determine the job finding rate?
- Typically, economist model it with a matching function that take vacancy and job search effort as the determinant for job finding (matching) rate
- Empirically, vacancy data has been survey by statistic agency (US in 2000 and Canada in 2015)
 - ▶ There are also works screening through online job posting website (e.g., indeed, monsters, etc)
- However, search effort is harder to measure
 - ▶ The CPS in the US and LFS in the Canada has supplement survey question asking about the job search effort
 - ▶ But researchers find different conclusion regarding its relationship to business cycle because of difference in method

Job Vacancy

- The vacancy fall when the economic hit a recession, and recover slowly afterward
- Particular high vacancy rate after the pandemic

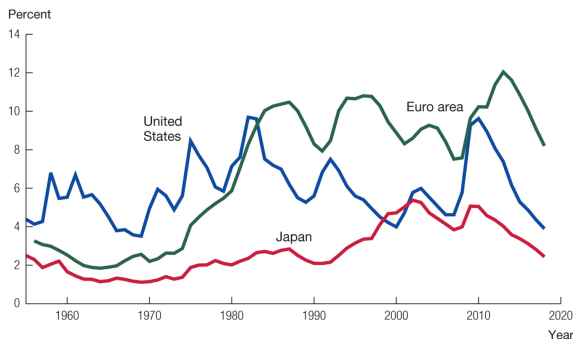


Labour Markets around the World

- Since 1980: Unemployment in Europe is well above the rate in the United States
- European unemployment has increased because of
 - ▶ Adverse shocks and high oil prices
 - ▶ Inefficient labour market institutions:
 - ★ Higher unemployment insurance
 - ★ Welfare benefits (high implicit tax rate for returning to work)

Unemployment in the United States, Europe, and Japan

- Japan's unemployment rate has been substantially below that in the United States and Europe until recent years
- European unemployment has been substantially higher than unemployment in the United States in recent decades
- Unemployment in Japan, after being low for many decades, began approaching U.S. levels in the 2000s



Source: OECD Main Economic Indicators and the U.S. Bureau of Labor Statistics.
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Hours Worked per Person

- When we look at labour market, we want to examine the intensive margin (hours worked) and extensive margin (employment)

TABLE 7.2

Hours Worked per Person (U.S. = 100 in 2017)

	1960	1990	2017
United States	89	105	100
Italy	109	89	85
France	109	79	75
Germany	133	94	85
U.K.	116	98	96
Japan	132	127	111
South Korea	81	135	127

Source: Penn World Tables, Version 9.1.
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Labour Markets around the World

- GDP per capita is lower in Europe. Why? People work fewer hours
- If working less is voluntary:
 - ▶ Europeans enjoy leisure or home production more
 - ▶ Welfare is likely improved
- If working less is due to distortions in the labour market:
 - ▶ This outcome is likely not welfare enhancing

What are You Going to Learn (Ch. 8)

- Understand the nature and costs of inflation. Explore the insights provided by the quantity
- Theory of money and the classical dichotomy into the phenomenon of inflation
- Examine the connection between interest rates and inflation using the Fisher equation
- Recognize the crucial association between fiscal policy and the occurrence of high inflation.

Inflation

- Inflation: Percentage change in the overall price level

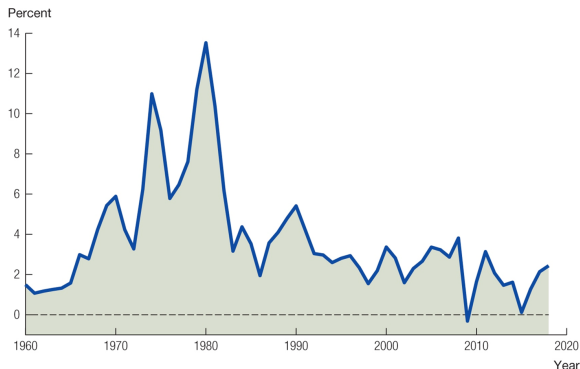
$$\text{Inflation rate} = \frac{P_{t+1} - P_t}{P_t} \times 100$$

where P_t is the price in period t

- Measured usually based on the Consumer Price Index (CPI)
 - ▶ CPI: Price index for a bundle of consumer goods

Inflation Rate in the United States

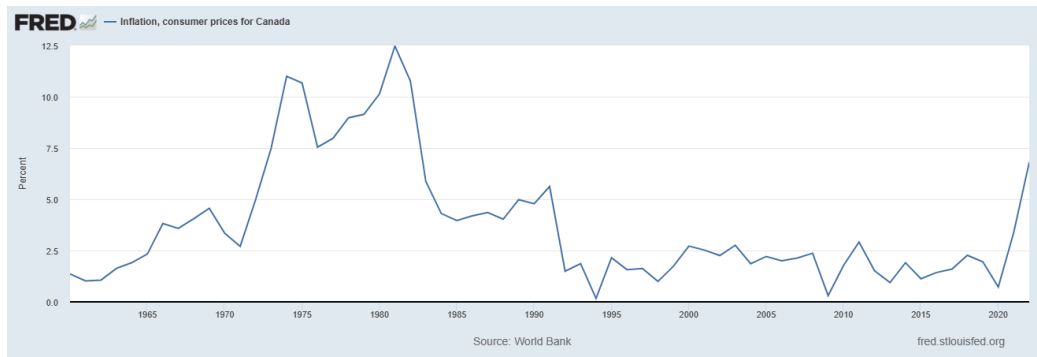
- The inflation was around 2% in the 60s and after the 80s
- However, inflation was highest in 1980 at 13.5 percent
 - ▶ High inflation created arbitrary transfers of wealth and distorted incentives for investment
 - ▶ Businesses that borrowed to finance investment faced extremely high interest rates



Source: World Bank, Inflation, consumer prices for the United States [FPCPITOTLZGUSA], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/FPCPITOTLZGUSA>, October 16, 2019.
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Inflation Rate in the Canada

- Canada's experience is similar to that in US
- Recent inflation surge is not unseen before



Case Study: How Much Is That?

- We can use the CPI to evaluate the value of a good in 1950 in today's dollars

TABLE 8.1

The Consumer Price Index, 1900–2018 (2018 = 100)

Year	CPI	Year	CPI
1900	3.24	1980	32.81
1930	6.65	1990	52.05
1950	9.59	2000	68.58
1960	11.80	2010	86.84
1970	15.47	2018	100.00

Source: Samuel H. Williamson, "The Annual Consumer Price Index for the United States, 1774–Present," MeasuringWorth, www.measuringworth.com.

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Case Study: How Much Is That?

- In 1950, a gallon of gasoline cost 27 cents, while in 2018 it cost around \$2.50 in the United States
- To compare, we can find the price of the good in 1950 in 2018 dollars

$$0.27 \times \frac{100}{9.59} = 2.82$$

- We can also compute the other way around

$$2.5 \times \frac{9.59}{100} = 0.24$$

- Real difference is not as large as the nominal
 - ▶ Gasoline was actually more expensive in 1950 than in 2018, contrary to how low the 27 cents a gallon sounds

What is Money?

- We often think of money as paper currency
- But, historically, money was backed by gold or if we go further back, seashells, rocks, other precious metals, etc
- Today, paper currency has value because of government decree (fiat money)
 - ▶ It is no longer backed by gold or silver
 - ▶ There is a mutual agreement that paper currency will be accepted as payment
 - ▶ Money has value because we expect others will value it and accept it as payment

The Quantity Equation

- Connects money/currency (M_t) and inflation (change in P_t)
- Velocity of money (V): the average number of times per year that each piece of paper currency is used in a transaction
- The amount of money used in purchases ($M_t V_t$) is equal to nominal GDP ($P_t Y_t$)

$$M_t V_t = P_t Y_t$$

The Classical Dichotomy and Constant Velocity

- The classical dichotomy: the real and nominal sides of the economy are completely separate in the long run
 - ▶ Economy 1 is produce 10 units of goods and charge \$50,000
 - ▶ Economy 2 is produce 10 units of goods and charge \$1,000,000
 - ▶ The two economy has the same real economy while the nominal side is different
 - ▶ In the growth model, the variables do not depend on money
- In the quantity theory of money:
 - ▶ Real GDP is assumed as exogenously given, determined by real forces (e.g., saving rate, etc)

$$Y_t = \bar{Y}_t$$

The Classical Dichotomy and Constant Velocity

- The velocity of money: exogenously given constant and assumed to be constant over time

$$V_t = \bar{V}$$

- ▶ the velocity of the M2 definition of money is approximately constant
- ▶ So, in the long run, this is not an unreasonable assumption
- The money supply: determined by the central bank and is exogenously given

$$M_t = \bar{M}_t$$

The Quantity Theory of Money

- The model has 4 Equations and 4 Unknowns

- The quantity equation

$$M_t V_t = P_t Y_t$$

- Real GDP from growth model

$$Y_t = \bar{Y}$$

- Exogenous and constant velocity

$$V_t = \bar{V}$$

- Exogenous money supply

$$M_t = \bar{M}$$

- Exogenous variables/parameters: $\bar{M}_t$, $\bar{V}$ and $\bar{Y}_t$

The Quantity Theory for the Price Level

- Plug in all the exogenous variables
- Solve for the price level:

$$P_t^* = \frac{\bar{M}_t \bar{V}}{\bar{Y}_t}$$

- Prices will rise as a result of
 - ▶ Increases in the money supply
 - ▶ Decreases in real GDP
- In the long run, the key determinant of the price level is the money supply (if Y_t is determined as in Solow model or Romer model)

The Quantity Theory for Inflation

- Inflation is the growth of the price, applying the rules of growth rates to the quantity equation

$$\bar{g}_M + \bar{g}_V = \bar{g}_P + \bar{g}_Y$$

- In the long run, V is not changing, thus $\bar{g}_V = 0$
- Let $g_P = \pi$

$$\pi^* = \bar{g}_M - \bar{g}_Y$$

The Quantity Theory for Inflation

$$\pi^* = \bar{g}_M - \bar{g}_Y$$

- Changes in money growth do not change real GDP because of the classical dichotomy
- So, the change in money lead to one-for-one to changes in the inflation rate
- Based on this model, Central bank has a lot of influence on the price level in the long run
- Empirically, this holds up both in U.S. and worldwide data

Measures of the Money Supply

- The monetary base includes currency and accounts (reserves)
 - ▶ Banks hold accounts with the economy's central bank
 - ▶ Banks ensure that they have cash on hand in case of withdrawals
- We add less liquid from C to MB to M1 to M2
- Most of the money are in form of savings deposits and individual money market accounts

TABLE 8.2

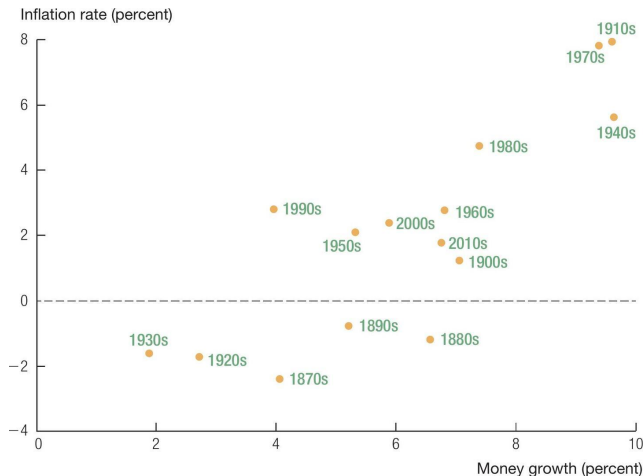
Different Measures of the Money Supply in January 2019
(billions of dollars)

C	Currency	1,710
MB	Monetary base = currency plus reserves	3,350
M1	Currency plus demand deposits (e.g., checking accounts)	3,740
M2	M1 plus savings deposits and individual money market accounts	14,470

Source: Federal Reserve Board of Governors, H.3 and H.6.
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Money Growth and Inflation in the United States, 1870–2018

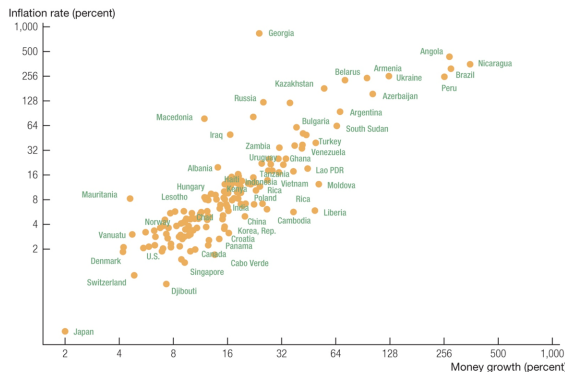
- A strong positive relationship between money growth and inflation



Sources: Money growth rates are computed using the data on M2 from the *Historical Statistics of the United States* (Cambridge: Cambridge University Press, 2006) and FRED. Inflation data are taken from measuringworth.com.

Money Growth and Inflation around the World, 1990–2017

- In the lower left corner, many countries have low money growth and low inflation (e.g., US, $\bar{g}_M = 5.4\%$ with $\pi = 2.8\%$ and Germany $\bar{g} = 2.8\%$ with $\pi = 1.9\%$)
- In the upper right corner, we have Peru, Brazil and so forth



Source: World Bank, *World Development Indicators*. The figure plots the average annual percentage change in the price level (inflation) and money supply for a sample of countries. Ratio scales are used for both axes.

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Revisiting the Classical Dichotomy

- When all prices in the economy double, relative prices are unchanged
- When the relative prices of goods are unchanged, nothing real is affected
 - ▶ If a country opts to introduce a new currency with two additional zeros at the end, the relative price of, say orange and apple, would remain the same
- The neutrality of money says that changes in the money supply have no real effects on the economy and affect only prices
- Empirically, it holds in the long run
- But, does not hold in the short run – nominal prices do not respond immediately to changes in the money supply

Real and Nominal Interest Rates

- The real interest rate is equal to the marginal product of capital and is paid in goods
- The nominal interest rate is the interest rate on a savings account and is paid in dollars

The Fisher Equation

- The Fisher equation:

$$i \approx R + \pi$$

where

i = nominal interest rate

R = real interest rate

π = inflation rate

- The nominal interest rate is generally high when inflation is high

The Fisher Equation

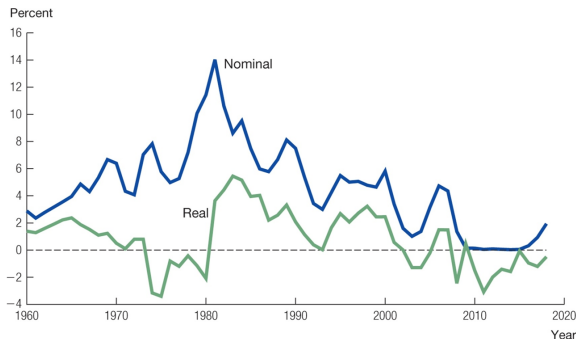
- We can rearrange the equation

$$R = i - \pi$$

- Empirically, the real interest rate has been negative
 - ▶ This implies that in the short run the real interest rate need not equal the MPK
- It is not a number that would be reported by the news or government
 - ▶ However, one should consider this metric when doing long-term investment
 - ▶ Say, someone retired on fixed income that give 10% return. However, if the person's cost of living increase by 10%, this retiree is losing purchasing power every month

Real and Nominal Interest Rates in the United States, 1960–2015

- The nominal interest rate is the highest in 1981 at 14 percent when the inflation was its highest in 1980 – This relationship is described by the Fisher equation
- In 2009, the real interest rate actually became larger than the nominal interest rate, due to deflation
- The real interest rate can also be negative, happened in the 1970s, 2003, and 2010–2015, when the nominal interest rate was very low



Source: FRED and author's calculations. The nominal interest rate plotted here is the annualized return on 3-month Treasury bills.

Costs of Inflation

- While the cost of inflation can apply to everyone in the economic, but not everyone face the same cost
- Individuals who are hurt during inflation:
 - ▶ An individual who has a pension that is not indexed to inflation
 - ▶ A bank that issues loans at fixed rates but that pays interest rates that move with the market
 - ▶ An individual with a variable rate mortgage

Costs of Inflation

- Low and stable inflation allows people and institutions to accurately predict inflation
 - ▶ They can then incorporate the predicted level of inflation in contracts
- Large surprise inflations can lead to large distributions in wealth
 - ▶ Borrowers are winners, as the real amount of debt falls
 - ★ Some firms may be able to get out from the debt in real term
 - ▶ Inflation is particularly costly when it occurs in an unexpected fashion and in an environment that is not prepared for the change

Costs of Inflation

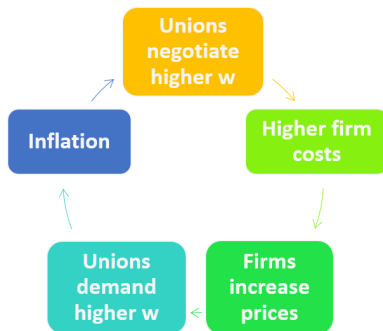
- Most countries experienced another cost of inflation as well, related to the interactions between inflation and the tax system
- In most countries, taxes are based on nominal income
- However, economic decisions based on real variables
- When inflation is high, your savings account may be earning a high nominal return that gets taxed, even though the real return is low
- Many government realized the problem and have the income tax schedule be indexed to inflation (partly)

Costs of Inflation

- Inflation distorts relative prices: some prices adjust quickly to inflation, while others adjust slowly
- Shoe leather costs of inflation
 - ▶ People want to hold less money when inflation is high, which means they must go to the bank more often
- Menu costs: the costs to firms of changing prices frequently

Case Study: The Wage-Price Spiral

- US faced rising inflation and relatively high unemployment in the early 70s
 - ▶ Inflation rise from 2% to 5% and unemployment 3.5% to 5%
- To maximize his chances for reelection, Nixon looked for a policy that would quickly address both problems...
 - ▶ Price control: freeze wages and prices at their current levels
 - ▶



The Fiscal Causes of High Inflation

- Inflation can be high if the central bank increases the money supply
- Knowing this, central bank would be restraint. However, they may “print” money on government’s behave to pay government’s bills
- The government’s uses of funds must equal its sources of funds

Government Funding = Tax Revenue + Borrowing + Changes in the stock of money

$$G = T + \Delta B + \Delta M$$

The Inflation Tax

- Seigniorage or the inflation tax: names for the revenue that the government obtains from printing more money (ΔM)
- The inflation tax will shows up as a rise in the price level and is paid by people holding currency
 - ▶ When government double the money supply, the price level would double in the long run according to the quantity theory of money
 - ▶ If each person in the economy holds \$200 in currency, the government is printing an extra \$200 for each person but then spends that currency itself
 - ▶ The real economic would not change, therefore, the \$200 of currency held by the each person is now worth only half as much in real terms

The Inflation Tax

- If a government runs large budget deficits ($G > T$), as debt rises ($\Delta B > 0$ for multiple periods),
 - ▶ lenders may worry the government will have trouble paying back loans and they may stop lending to the government altogether or charge a higher interest rate
 - ▶ In 1990s, Canada faced a crisis as such
- Debt solution: Raising taxes? May not be politically feasible
- The government may resort to printing currency to finance its budget
 - ▶ Lenders to the government will be paid back in currency that is worth less than the dollars lent
 - ▶ Knowing this, the debt could be in foreign currency for some developing countries, like in USD or Euro

Central Bank Independence

- There is temptation to print money to pay for its spending is there for all governments
- To avoid this temptation, many countries establish a kind of separation between the central bank and the government
 - ▶ i.e., the federal government cannot order the central bank to issue more currency so that the government can pay its bills
- In US and Canada, the central bank are the Federal Reserve and Bank of Canada
- In much of Western Europe, these decisions are even more sharply separated:
 - ▶ Each country that uses the euro has its own government, which is responsible for spending and taxation
 - ▶ Monetary policy is conducted by a multinational European Central Bank
- Central bank independence is seen as an attempt to prevent fiscal considerations from leading to excessive inflation or political cycle

Case Study: Episodes of High Inflation

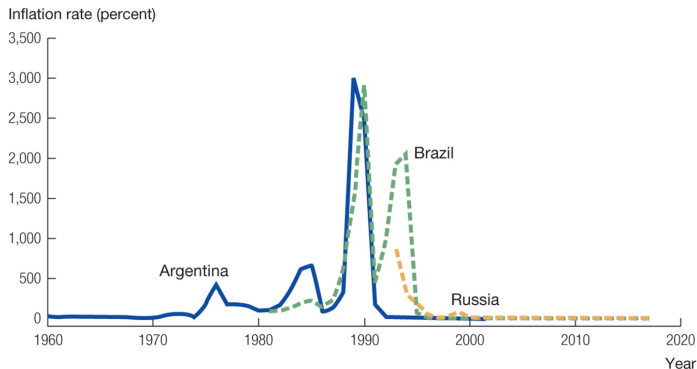
- Hyperinflation: episodes of extremely high inflation
 - ▶ Greater than ≈ 500 percent per year, has occurred throughout history
- Examples of hyperinflation:
 - ▶ In the 1980s and 1990s, numerous episodes among the economies of Latin America
 - ▶ The inflation rate in Russia in the early 1990s transitioning to a noncommunist country reached more than 800 percent per year
 - ▶ Zimbabwe in November 2008: prices doubled roughly every 24 hours and the inflation rate peaked at nearly 80 billion percent per month
 - ▶ From 1919 to 1923, prices in Germany rose by over a factor of a trillion and were rising 300-fold each month
- Countries experiencing hyperinflation typically raise about 5 percent of GDP from the inflation tax (e.g., Argentina raised 10 percent of GDP this way)

Case Study: Episodes of High Inflation

- Episodes of high inflation tend to recur
- How do episodes of high inflation or hyperinflation end?
 - ▶ The answer from the quantity theory is that they end when the rate of money growth falls sharply
 - ▶ The more fundamental analysis of the government's budget constraint, however, reveals that this can occur only when the government makes difficult choices to get its finances in order:
 - ★ Lower spending
 - ★ Higher taxes
 - ★ New loans (usually from IMF, with very tough requirement)

Hyperinflations in Argentina, Brazil, and Russia

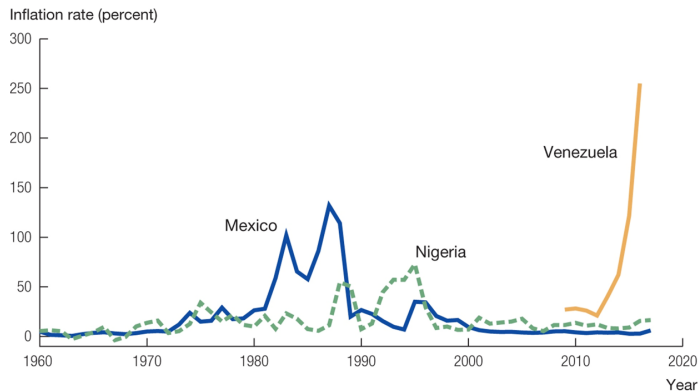
- Episodes of high inflation tend to recur In Argentina and Brazil
- Hyperinflations can stop quickly



Source: FRED.
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High Inflation in Mexico, Nigeria, and Venezuela

- Mexico and Nigeria experienced high inflation, but not what we would call hyperinflation.
- Venezuela is now suffering from hyperinflation



Source: FRED.
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Case Study: Episodes of High Inflation

- Hyperinflation ends when the rate of money growth falls rapidly and when the government gets its finances in order through lower spending, higher taxes, and new loans
- Another difficulty in ending high inflation is the coordination problem
 - ▶ Imagine you are a business owner and you have to choose how to set your prices in the coming week
 - ▶ If the rate of inflation has been high, people build their expectations into the prices they set
 - ▶ This coordination injects a certain inertia into the inflation process
 - ▶ At some level, all the price setters in the economy have to be convinced that the high inflation of recent years is going to end, and this coordination problem can be hard to solve
 - ▶ Because fiscal reform involves difficult choices and because of the coordination problem, halting hyperinflations is not easy

Next Lecture

- Cover Ch. 17 and part of Ch.7/8